

Working Title

YT

September 30, 2026

Abstract

Introduction
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1 Introduction

the time of symptom onset determines access to treatment. (stroke treatment guidelines)
study showed that unknown-onset stroke is common, distinct from wake-up stroke, and associated with lower reperfusion treatment, and poorer outcomes [1].

Main Aim:

The primary aim of this study was to examine temporal and geographic variation in uncertain symptom-onset stroke among stroke patients in Norway.

2 Methods

Data from the norwegian stroke registry, statistics norway

fom 2014 - tom 2013

inklusion criteria: I61 - hjerneblødning I63 - hjerneinfarkt I64 - hjerneslag, ikke spesifisert som infarkt eller blødning

nødvendig med sikre akutte fokale nevrologiske utfall

for bevistløse pasienter: akutte fokale nevrologiske utfall med positiv bildediagnostikk

innlagt innen 28 døgn etter symptomdebut (nytt slag etter 7 døgn registreres som nytt tilfelle, men ikke de innen 7 døgn)

ved trombolyse der det ikke blir sekvele eller bildefunn etter innleggels: kodes som I 63.

Pasienter med normal kontroll MR etter trombolytisk behandling skal kodes som hjerneinfarkt i ICD10.

Flow chart

88228

Innlagt i sykehus: 4571

83657

Våknet med symptom: 14757

68900

komnr missing: 38

68862

Table 1: Characteristics of patients

	Uncertain onset (n = 12,612)	Other stroke (n = 55,233)	SMD
Age, mean (SD)	73.89 (13.4)	73.31 (13.4)	0.043
Female	5,945 (47.1)	25,337 (45.9)	0.025
Stroke type, Infarction	10,813 (85.7)	47,360 (85.7)	0.009
Previous TIA	880 (7.0)	4,356 (7.9)	0.043
Previous MI	1,482 (11.8)	6,816 (12.3)	0.020
Atrial fibrillation	2,917 (23.1)	13,119 (23.8)	0.015
Diabetes	2,438 (19.3)	9,234 (16.7)	0.069
Antihypertensive medication	6,508 (51.6)	28,464 (51.5)	0.040
Lipid-lowering medication	3,700 (29.3)	16,821 (30.5)	0.050
Thrombolysis	161 (1.3)	12,148 (22.0)	0.685
Smoking status			0.086
Never smoker	4,673 (37.1)	20,966 (38.0)	
Current smoker	2,717 (21.5)	10,234 (18.5)	
Former smoker	2,998 (23.8)	14,538 (26.3)	
Unknown	2,224 (17.6)	9,495 (17.2)	
Level of Consciousness (admission)			0.087
Alert	10,856 (86.1)	47,236 (85.5)	
Mildly Drowsy	800 (6.3)	4,283 (7.8)	
Markedly Drowsy	349 (2.8)	1,617 (2.9)	
Unresponsive	432 (3.4)	1,699 (3.1)	
Unknown	175 (1.4)	398 (0.7)	
Pre-stroke modified Rankin Scale			0.098
No symptoms	6,884 (54.6)	31,923 (57.8)	
No significant disability	1,927 (15.3)	8,805 (15.9)	
Slight disability	1,369 (10.9)	5,319 (9.6)	
Moderate disability	931 (7.4)	3,829 (6.9)	
Moderately severe disability	385 (3.1)	1,612 (2.9)	
Severe disability	54 (0.4)	227 (0.4)	
Unknown	1,062 (8.4)	3,518 (6.4)	
Living situation			0.173
Living with others	6,192 (49.1)	31,614 (57.2)	
Living alone	5,701 (45.2)	20,548 (37.2)	
Institution/nursing home	376 (3.0)	1,490 (2.7)	
Unknown	333 (2.8)	1,581 (2.8)	
Education			0.055
No education	60 (0.5)	203 (0.4)	
Compulsory education	4,150 (33.2)	17,149 (31.3)	
Upper-secondary education	5,868 (46.9)	25,743 (47.0)	
Higher education	2,419 (19.3)	11,606 (21.2)	
Unknown	15 (0.1)	60 (0.1)	
Household income			0.068
Low	3,325 (26.4)	13,437 (24.3)	
M. low	3,138 (24.9)	13,278 (24.0)	
M. high	3,009 (23.9)	13,470 (24.4)	
High	3,092 (24.5)	14,885 (26.9)	
Unknown	48 (0.4)	163 (0.3)	

SMD = standardized mean difference; SD = standard deviation; TIA = transient ischaemic attack.

2.1 Statistical analysis

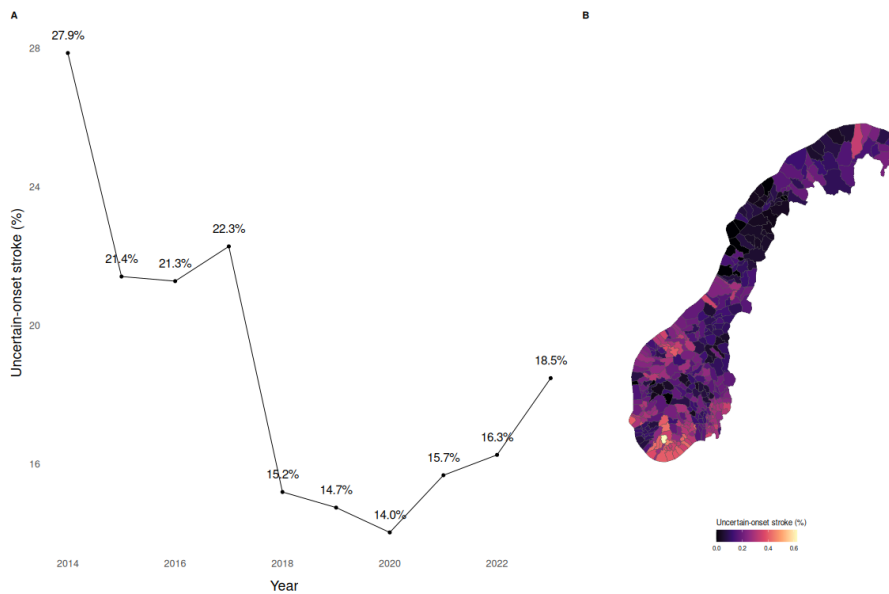
3 Results

The study population included 67,845 patients with stroke registered in the Norwegian Stroke Registry between 2014 and 2023, of whom 12,612 (18.6%) had uncertain-onset stroke (table 1). Most patient characteristics were similar between patients with uncertain-onset stroke and all other strokes, with standardized mean differences generally below 0.10. The largest differences were observed for living situation, and thrombolysis, while differences in demographic characteristics, stroke type, and cardiovascular history were modest. Living alone was more common among patients with uncertain-onset stroke (45.2% vs 37.2%), and they recieved thrombolysis less frequently (1.3% vs 22.0%).

Temporal and geographic variation in uncertain-onset stroke

The crude proportion of uncertain-onset stroke varied over time, with the heigest proportion in 2014 (27.9%) and the lowest in 2020 (14.0%). The proportion declined during the early years of the study period, with a marked reduciton between 2017 and 2018. It remained relatively low through 2020, followed by a gradual increase to 18.5% in 2023 (figure 1A).

Figure 1: Temporal and geographic variation in uncertain-onset stroke. A) Annual proportion of patients with uncertain-onset stroke, 2014–2023. B) Municipality-level geographic variation in uncertain-onset stroke.



In addition to the temporal variation, substantial geographic variation in uncertain-onset stroke was observed across municipalities, ranging from 0% to 62% with a median of 16.7% (IQR 10.2% and 25.8%) (Figure 1B). Areas with relatively high and low proportions appeared to cluster together geographically rather than an even distribution across the country.

Table 2: Factors associated with uncertain-onset stroke

	Predictor	OR (95% CrI)
Year	2015 vs 2014	0.67 (0.61–0.73)
	2016 vs 2014	0.64 (0.59–0.70)
	2017 vs 2014	0.63 (0.58–0.69)
	2018 vs 2014	0.37 (0.34–0.41)
	2019 vs 2014	0.36 (0.33–0.40)
	2020 vs 2014	0.24 (0.22–0.27)
	2021 vs 2014	0.28 (0.25–0.31)
	2022 vs 2014	0.30 (0.28–0.34)
	2023 vs 2014	0.41 (0.37–0.45)
Living situation	Living alone	1.19 (1.13–1.25)
	Institution/nursing home	0.80 (0.70–0.92)
Smoking	Current smoker	1.11 (1.04–1.18)
	Former smoker	0.95 (0.90–1.01)
Gender	Female	0.95 (0.90–0.99)
Income	Second quartile	0.98 (0.92–1.04)
	Third quartile	0.96 (0.89–1.02)
	Highest quartile	1.00 (0.93–1.07)
Education	Compulsory education	1.03 (0.85–1.26)
	Secondary education	1.04 (0.86–1.27)
	Higher education	1.07 (0.87–1.31)
Previous TIA	No	1.10 (1.01–1.20)
Previous MI	No	1.05 (0.98–1.13)
Atrial fibrillation	No	1.09 (1.03–1.15)
Diabetes	No	0.86 (0.81–0.91)
Antihypertensives	No	1.00 (0.96–1.05)
Lipid-lowering treatment	No	1.03 (0.98–1.09)
Pre-stroke mRS	mRS 1	1.03 (0.97–1.10)
	mRS 2	1.13 (1.05–1.23)
	mRS 3	1.11 (1.01–1.22)
	mRS 4	1.17 (1.02–1.35)
	mRS 5	1.41 (1.00–2.00)
Stroke type	Infarction	0.98 (0.91–1.04)
Consciousness	Mildly drowsy	0.72 (0.66–0.79)
	Markedly drowsy	0.78 (0.68–0.90)
	Unresponsive	0.89 (0.78–1.01)
wake-up-status Missingness	Individual	6.68 (6.36–7.02)
	Municipality-year	5.30 (4.29–6.56)

TIA = transient ischaemic attack; MI = myocardial infarction; mRS = modified Rankin Scale. Bold estimates indicate 95% credible intervals excluding 1. The model additionally included municipality effects, age, and health trust.

Table 3: Comparison of spatial models for uncertain-onset stroke

Model	Spatial SD	95% CrI	ϕ	DIC / WAIC
Unadjusted	0.696	0.614–0.781	0.976	61088.99 / 61089.08
Sociodemographic	0.697	0.616–0.783	0.976	60753.25 / 60752.12
Clinical	0.700	0.618–0.786	0.975	60780.37 / 60782.12
Missingness	0.775	0.683–0.871	0.969	54270.76 / 54269.53
Health trust	0.518	0.444–0.595	0.970	60710.35 / 60710.54
Full model	0.612	0.529–0.670	0.969	53638.95 / 53640.24

Spatial SD - the posterior standard deviation of the spatial effect. DIC = deviance information criterion; WAIC = Watanabe-Akaike information criterion. ϕ - the proportion of spatial variance attributed to the structured component.

unadjusted and unadjusted pairs from 2014 to 2023

4 Discussion

This is a placeholder Discussion section added to verify the LaTeX build and editor setup. Replace this text with interpretation of results and context.

Supplementary

References

- [1] M Søyland et al. “Wake-up stroke and unknown-onset stroke; occurrence and characteristics from the nationwide Norwegian Stroke Register”. In: *European Stroke Journal* 7.2 (2022), pp. 143–150. DOI: 10.1177/23969873221089800.

